LABORATORY 2

MOS DEVICE CHARACTERIZATION

OBJECTIVES

No.	Objectives	Requirements
1	Characterize the threshold voltage	▪ Determine the threshold voltage and operation region of the NMOS and PMOS device using $I_{DS} = f(V_{GS})$
2	The effects on I/V characteristics when V_{GS}, width, and length vary of NMOS transistor	▪ Drawing curves $I_{DS} = f(V_{DS})$ when these parameters vary. ▪ Consider curves drawn.
3	The effects on I/V characteristics when V_{SG}, width, and length vary of PMOS transistor	▪ Drawing curves $I_{SD} = f(V_{SD})$ when these parameters vary. ▪ Consider curves drawn.
4	Measure input gate capacitance of MOS device	▪ Measure input gate capacitance of NMOS/PMOS devices and compare

LAB 2 INFORMATION

- Consider ***NMOS nmos1v_lvt device, PMOS pmos1v_lvt device*** in ***gpdk45nm*** and power supply $\boldsymbol{vdd = 1.0V}$
- **Note:** All figures in the document are for reference only.

EXPERIMENT 1

1.1. Objective: Measure and characterize threshold voltage.

1.2. Requirements: Characterize and measure the threshold voltage V_{TH} using two methods. Investigate the VTH of an NMOS transistor and PMOS transistor while varying the W, L, and PVT (Process, Voltage, Temperature) conditions.

- *Method 1:* Observe waveform method
- *Method 2:* Extraction from the DC operating point.

1.3. Instruction: Assemble the circuit as shown in ***Figure 1 and Figure 2***, setting parameters for power supplies (using ***vdc*** source), with L = $45nm$, W = $120nm$, $V_{SB} = V_{BS} = 0V$.

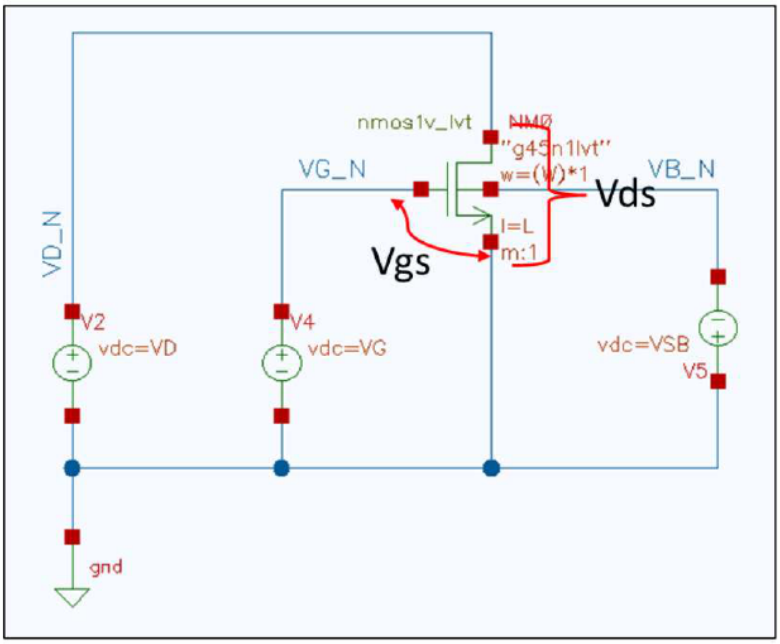


Figure 1. Circuit diagram for characterizing NMOS characteristics

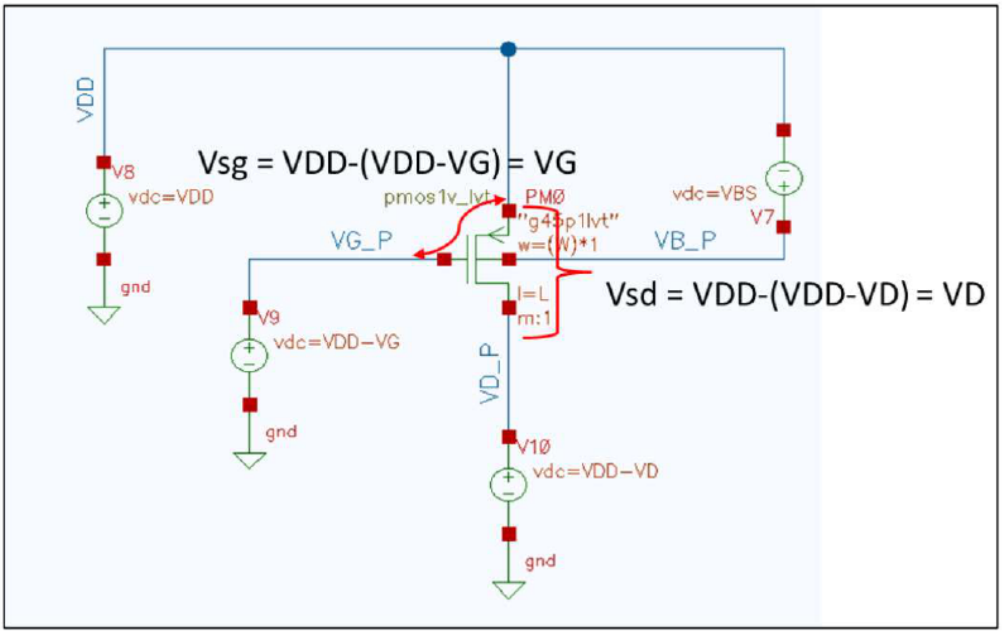
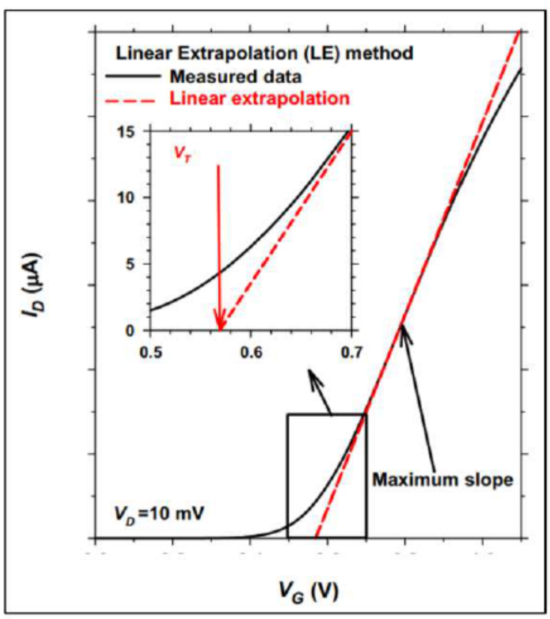


Figure 2. Circuit diagram for characterizing PMOS characteristics

1.4. Extract Threshold Voltage

1.4.1. Observe waveform method

- Propose ***at least one method*** to determine the threshold voltage of NMOS ***nmos1v_lvt*** and PMOS ***pmos1v_lvt*** devices. Please refer to the reference document for further details.



Source: Revisiting MOSFET threshold voltage extraction methods

a. Testcase

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25^oC
W	$120nm$
L	$45nm$
$\boldsymbol{V_{DD}}$	$1\ V$
$\boldsymbol{V_{GS}}$	$\mathbf{0: 0.01: 1\ V}$
$\boldsymbol{V_{SB}}$	$0\ V$
$\boldsymbol{V_{DS}}$	$0.8V$
Observe	$\boldsymbol{I_{DS}}$ **with respect to** $\boldsymbol{V_{GS}}$

b. Waveform

- Draw a curve when fixing the value $V_{DNS} = V_{SDP} = 0.8V, V_{SBN} = V_{BSP} = 0V$ and sweeping variable $V_{GSN} = V_{SGP}$ from 0V to 1.0V with step = 0.01V to characterize the change of I_{DS} with respect to V_{GS}/V_{SG}.
- **Determine the threshold voltage of NMOS and PMOS from $\boldsymbol{I_{DS} = f(V_{GS})}$ as Figure 3.**

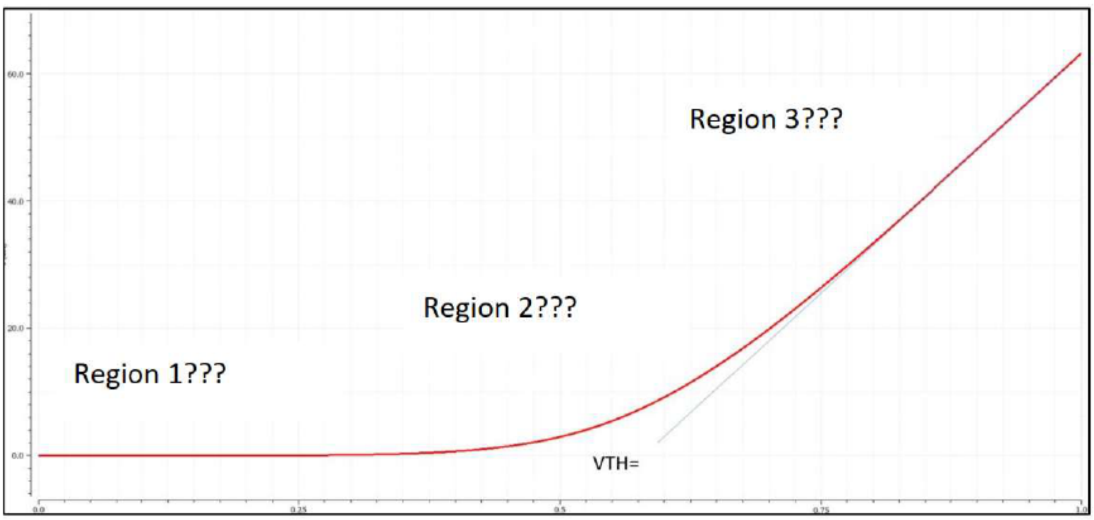


Figure 3. Waveform NMOS $I_{DS} = f(V_{GS})$

c. Observe from the waveform

Item	NMOS	PMOS
VTH		
Region 1	**VGS = ...** **VDS = ...**	**VGS = ...** **VDS = ...**

	Vov = ... **VGS ... VTH** **VDS ... Vov** **=> Region ...**	**Vov = ...** **VGS ... VTH** **VDS ... Vov** **=> Region ...**
Region 2	**VGS = ...** **VDS = ...** **Vov = ...** **VGS ... VTH** **VDS ... Vov** **=> Region ...**	**VGS = ...** **VDS = ...** **Vov = ...** **VGS ... VTH** **VDS ... Vov** **=> Region ...**
Region 3	**VGS = ...** **VDS = ...** **Vov = ...** **VGS ... VTH** **VDS ... Vov** **=> Region ...**	**VGS = ...** **VDS = ...** **Vov = ...** **VGS ... VTH** **VDS ... Vov** **=> Region ...**

d. *Explanation (if applicable).*

1.4.2. Extract from the DC operating point method

Outputs Setup | Results | Run Preview

Name	Type	Details	Plot
VTH_P	expr	OP("/PM0" "vth")	✔

- Write the calculator equation to measure VTH of NMOS and PMOS.
- Compare the threshold voltage between 2 methods.

	Observe waveform method	Extract from the DC operating point method
VTH NMOS (mV)		
VTH PMOS (mV)		

1.5. Threshold voltage characterization

- Using method "***Extraction from the DC operating point***", measure V_{TH} in testcases below.
- Use the testbench as **Figure 1** and **Figure 2** to measure the V_{TH} while varying the W, L, and PVT (Process, Voltage, Temperature) conditions.
- **Collect data and summary the result.**

Parameter	Testcase a Vary W	Testcase b Vary L	Testcase c Vary Temp	Testcase d Vary Process	Testcase e/f Vary VGS/VDS	Testcase j Vary VSB
Device	nmos1v_lvt					
Process	TT			**SS, TT, FF**	TT	
Temp	25^oC		$\mathbf{-40, 25, 125^oC}$	25^oC		
W	***250n, 500n, 750n, 1000nm***	120nm				
L	$45nm$	***45nm,90n, 180n,270nm***	45nm			
$\boldsymbol{V_{DD}}$	$1\,V$					
$\boldsymbol{V_{GS}}$	$0.8V$				**e: 0.1,0.5,0.8V**	
$\boldsymbol{V_{SB}}$	$0\,V$					**-0.5, -0.2, 0.0, 0.2, 0.5V**
$\boldsymbol{V_{DS}}$	$0.8V$				**f: 0.1,0.5,0.8V**	
Observe	$\boldsymbol{V_{TH}}$ **from DC operating point**					

- ***Example table:***

Testcase 1: Vary Width

Width (nm)	250	500	750	1000
VT_H NMOS (mV)				
V_{TH} PMOS (mV)				

Testcase 2: Vary Length

Length (nm)	45	90	180	270
V_{TH} NMOS (mV)				
V_{TH} PMOS (mV)				

...

- **Explain the result if applicable.**

1.6. Plot and measure Trans-conduction (g_m)

- Plot g_m curve from $I_D = f(V_{GS})$

$$g_m = \frac{\Delta I_D}{\Delta \mathrm{V_{GS}}}$$

- Measure the g_m at linear and saturation region

EXPERIMENT 2

2.1. Objective: Known NMOS operations and I-V characteristics

2.2. Requirements: Simulate and draw curves $I_{DS} = f(V_{DS})$ of NMOS and PMOS currently used.

2.3. Analyze: Your report must show these results and **explain the NMOS operation region based on waveform and theory**.

- *2.3.1. Testcase*

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25^oC
W	$120nm$
L	$45nm$
V_{DD}	$1\ V$
V_{GS}	$0.8\ V$
V_{SB}	$0\ V$
V_{DS}	$0: 0.01: 1V$
Observe	I_{DS} **with respect to** V_{DS}

- *2.3.2. Waveform*

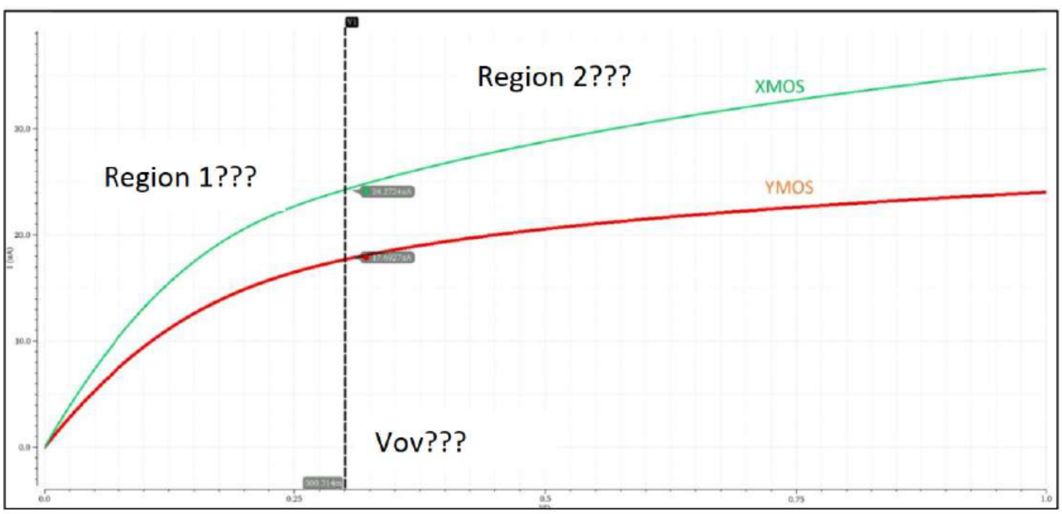


Figure 4. Waveform PMOS & NMOS $|I_{DS}| = f(|V_{DS}|)$ example

- *2.3.3. Observe from the waveform*

- Collect the result of NMOS and PMOS as the following table.

- Which curve corresponds to XMOS and which corresponds to YMOS—NMOS or PMOS?

Item	NMOS	PMOS
VTH		
Region 1	VGS = … VDS = … Vov = … VGS … VTH VDS … Vov => Region …	VGS = … VDS = … Vov = … VGS … VTH VDS … Vov => Region …
Region 2	VGS = … VDS = … Vov = … VGS … VTH VDS … Vov => Region …	VGS = … VDS = … Vov = … VGS … VTH VDS … Vov => Region …

➢ *2.3.4 Explanation (if applicable)*

- …

2.4. Measure channel length modulation parameter

- Propose method to measure channel length modulation parameter (λ) based on $I_D = f(V_{DS})$ waveform.

EXPERIMENT 3

3.1. Objective: The effects on I/V characteristics of NMOS when V_{GS}, width and length vary.

3.2. Requirements: Simulate and draw curves when V_{GS}, width and length vary.

3.3. Instruction:

- Setting parameters *W*, *L*, V_{GS}, V_{SB} and V_{DS} for convenience when characterizing the circuit shown in **Figure 1**.
- Sweep multiple variables in parallel using ADE-Explorer: You can refer to the setting in **Figure 5** (don't follow completely the setup), and observe the result shown in **Figures 6** and **Figures 7**.

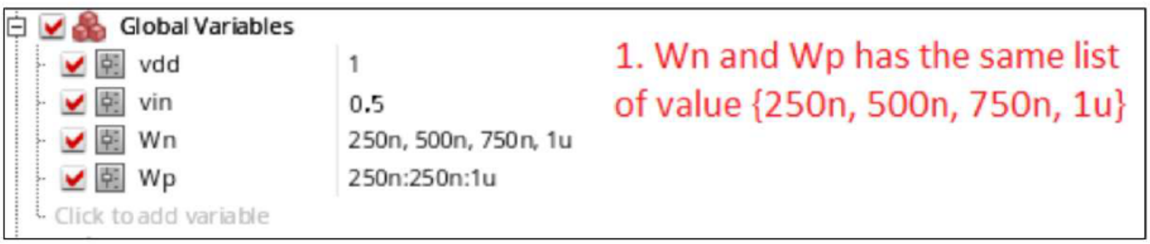


Figure 5. Setting parameters for changing W of NMOS $\{250, 500, 750, 1000\}\ nm$

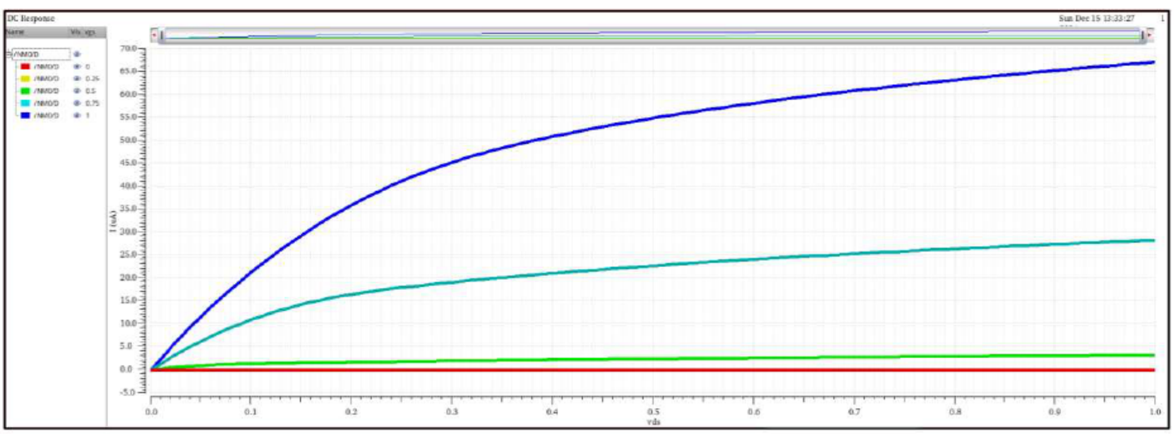


Figure 6. $I_{DS} = f(V_{DS})$ waveform with $V_{GS} = \{0, 0.25, 0.5, 0.75, 1.0\}\ V$

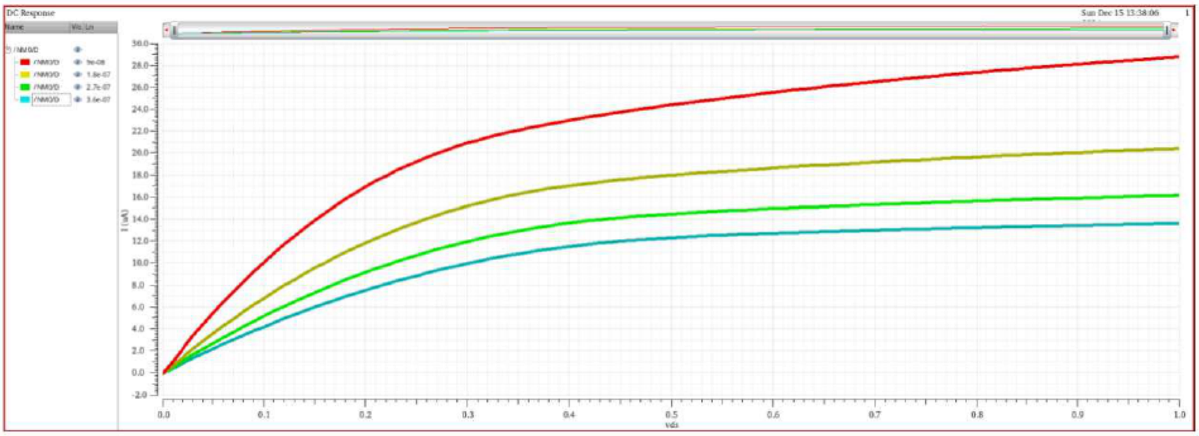


Figure 7. $I_{DS} = f(V_{DS})$ waveform with $L = \{90,180,270,360\}\ nm$

3.4. Analyze: Your result must show these results, **explain and summarize the conclusion of the current and operation region of NMOS with the width, length, and V_{GS} variation based on waveform and theory.**

- Choose $L = 45nm, W = 120nm$ and $V_{SB} = 0V$. Draw curves $I_{DS} = f(V_{DS})$, sweeping variable V_{DS} from 0V to 1.0V with step = 0.01V and sweeping V_{GS} from 0.0V to 1.0V with step = 0.25V.
- Choose $L = 45nm, V_{GS} = 0.8V$ and $V_{SB} = 0V$. Draw curves $I_{DS} = f(V_{DS})$ sweeping variable V_{DS} from 0V to 1.0V with step = 0.01V and $W = \{250, 500, 750, 1000\}\ nm$
- Choose $W = 500nm$, $V_{GS} = 0.8V$ and $V_{SB} = 0V$. Draw curves $I_{DS} = f(V_{DS})$ sweeping variable V_{DS} from 0V to 1.0V with step = 0.01V and $L = \{90,180,270,360\}\ nm$

EXPERIMENT 4

4.1. Objective: The effects on I/V characteristics of PMOS when V_{SG}, width and length vary.

4.2. Requirements: Know how to **create a testbench and consider the I/V characteristics of PMOS**.

4.3. Instruction:

- Setting parameters *W*, *L*, V_{SG}, V_{BS} and V_{SD} for convenience when characterizing the circuit shown in **Figure 2**.

4.4. Analyze: Your result must show these results, **explain and summarize the conclusion of the current and operation region of PMOS with the width, length, and V_{SG} variation based on waveform and theory.**

- Choose $L = 45nm, W = 120nm$ and $V_{BS} = 0V$. Draw curves $I_{SD} = f(V_{SD})$, sweeping variable V_{SD} from 0V to 1.0V with step = 0.01V and sweeping V_{SG} from 0.0V to 1.0V with step = 0.25V.
- Choose $L = 45nm, V_{SG} = 0.8V$ and $V_{BS} = 0V$. Draw curves $I_{SD} = f(V_{SD})$ sweeping variable V_{SD} from 0V to 1.0V with step = 0.01V and $W = \{250, 500, 750, 1000\}\ nm$.
- Choose $W = 500nm$, $V_{SG} = 0.8V$ and $V_{BS} = 0V$. Draw curves $I_{SD} = f(V_{SD})$ sweeping variable V_{SD} from 0V to 1.0V with step = 0.01V and $L = \{90,180,270,360\}\ nm$.

EXPERIMENT 5

5.1. Objective: Measure the gate capacitance of MOS device.

5.2. Requirements: Measure and compare the **gate capacitance** of **NMOS and PMOS transistors**

5.3. Instruction:

- Create separate testbench schematics for the **NMOS** and **PMOS** devices.
- **Bias each transistor in the appropriate operating region as specified by the instructor.**
    - Triode region: $V_{GSN} = V_{SGP} = 0.8V, V_{DSN} = V_{SDP} = 0.2V$
    - Saturation region: $V_{GSN} = V_{SGP} = 0.8V, V_{DSN} = V_{SDP} = 0.8V$
- Apply a small-signal AC or transient excitation to the **gate terminal** while properly biasing the source, drain, and body.
- Measure the effective **gate capacitance** (including C_{gs}, C_{gd} and overlap capacitances) from simulation results.
- **Repeat the measurement for different transistor sizes**
    - L = 45nm
    - W = 120n, 150n, 240n, 300n
- **Record and compare the measured gate capacitance values for NMOS and PMOS devices.**
- **Discuss the impact of device sizing and bias conditions on the measured gate capacitance.**

❖ Wrap-up Questions – Lab 2: Investigation of CMOS Characteristics and Operation

1. Why does the pinch-off point shift toward the drain in saturation region, and why does the inversion layer under the oxide disappear while current still flows from drain to source for NMOS?

→ Provide a physical explanation for channel formation and carrier movement under high V_{DS} conditions.

2. Why does the threshold voltage V_{TH} increase when the source-bulk voltage V_{SB} increases for NMOS?

→ Explain this phenomenon based on semiconductor physics (e.g. depletion width changes), rather than relying solely on the formula:

$$V_{TH} = \mathrm{V_{TH0}} + \gamma \sqrt{|2\Phi_F + V_{SB}| - |2\Phi_F|}$$

3. How can two short-channel transistors connected in series be modeled as one long-channel transistor?

→ Demonstrate this by analyzing the behavior of the stacked structure under different biasing conditions.

(Hint: Consider one transistor in saturation and the other in triode region to mimic a long channel's behavior.)